



Beyond the “source-sink” paradigm: catalytic mechanism of low-angle tilt grain boundaries in vacancy generation and annihilation



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Grain boundary (GB)-vacancy interactions play a pivotal role in tailoring material properties. However, the prevailing paradigm that “GBs function as vacancy sources/sinks” lacks robust atomic-scale evidence, thereby failing to discriminate the intrinsic mechanisms governing vacancy dynamics and leaving critical unresolved issues—most notably the GB capacity for vacancy generation/annihilation. To fill this gap, we adopt molecular dynamics simulations to systematically observe the vacancy thermal equilibration (VTE) process, wherein low-angle tilt GBs (LATGBs) are engineered as the exclusive vacancy sources and sinks. Our results unveil a remarkable phenomenon: LATGBs maintain invariant atomic structures despite continuous massive vacancy absorption and emission throughout VTE. Accordingly, we propose that LATGBs function as catalysts for vacancy generation/annihilation, with unlimited vacancy absorption/emission capacity. Furthermore, we corroborate four critical atomic-scale findings: (1) Vacancy generation/annihilation rate at edge dislocations reaches $8.62 \text{ ns}^{-1} \cdot \text{nm}^{-1}$; (2) Atomic migration rate within dislocation vacant channels is at least three orders of magnitude faster than grain interior diffusion; (3) Seven of eight sites surrounding each vacant channel are catalytically active due to ultralow vacancy formation energies; (4) Compressive-tensile stress interfaces along dislocations mediate vacancy migration between active sites and grain interiors. These atomic-scale insights establish a complete catalytic cycle governing LATGB-mediated vacancy generation/annihilation.

Grain boundaries (GBs) are ubiquitous features of polycrystalline materials. It is well established^{1–3} that their profound modulation effect on key material properties stems from their interactions with other defects. Among these, GB-vacancy interactions have garnered particular attention, as they dictate critical behaviors—such as diffusion, phase transformation, and plastic deformation—while being closely coupled to structural stability, mechanical strength, and damage tolerance.

The current dominant paradigm—that GBs act as vacancy sources/sinks—traces back to the 1950s, anchored by two seminal observations: Barnes et al.⁴ observed preferential precipitation of helium bubbles at GBs in irradiated copper following helium ion implantation, while Burke⁵ found that pores adjacent to GBs exhibited more rapid shrinkage than those within grain interiors during sintering. Accordingly, the authors hypothesized that GBs function as vacancy sources to supply vacancies for helium bubble

nucleation and growth, and as vacancy sinks to absorb excess vacancies and facilitate the shrinkage of nearby pores. Since then, numerous experimental studies^{6–11} have documented the generation and annihilation of vacancies at GBs, validating the source/sink paradigm from a phenomenological perspective. To date, this framework has remained well-entrenched and largely unchallenged in the academic community.

However, a comprehensive, definitive elucidation of the GB-vacancy interaction mechanism depends intrinsically on detailed atomic-scale insights—a goal that remains a formidable challenge for state-of-the-art experimental techniques¹². Thus, virtually all existing atomic-scale mechanistic investigations of GB-vacancy interactions resort to computer simulations, with employed methodologies including but not limited to molecular dynamics (MD)^{13–16}, density functional theory^{17–19}, Monte Carlo^{20,21}, kinetic Monte Carlo^{22,23}, and recent machine learning-based

simulations^{24–26}. To date, all such computer simulations exhibit two key characteristics: (1) tailoring GB structures via the artificial removal or addition of atoms to mimic the GB structural evolution arising from vacancy absorption or emission, with the aim of clarifying the underlying interaction mechanisms; (2) elucidating the properties of static defects within GBs or probing the local migration of isolated defects to and from GBs.

Against this backdrop, a critical fundamental question arises: **by what mechanisms, and to what extent**, do GBs mediate vacancy generation and annihilation in their purported role as the vacancy sources and sinks? On this topic, existing studies have focused exclusively on tilt GBs and have primarily addressed the “mechanism” component of this question. The climb theory is rooted in the dislocation model of tilt GBs²⁷—which conceptualizes tilt GBs as ordered walls of parallel edge dislocations—and posits that tilt GBs absorb or emit vacancies via the climb of GB dislocations^{28–33}. The phase transition theory indicates that tilt GBs mediate these processes through two-dimensional phase transitions^{15,34,35}, in which varying GB vacancy concentrations trigger reversible structural transformations. Using MD simulations of cascade collisions in the vicinity of GBs, Bai et al.¹⁴ proposed an interstitial theory, whereby tilt GBs preferentially trap self-interstitial atoms and annihilate nearby vacancies by emitting these interstitials, rather than directly absorbing the vacancies. Thus far, it is unequivocally established that GBs act as both sources and sinks for vacancies. This consensus aligns with experimental observations and theoretical frameworks elucidating the underlying mechanisms, albeit with the envisaged mechanisms remaining distinctly divergent. Furthermore, the “extent” component of the question remains unaddressed. When functioning as a source, a GB inherently possesses an upper limit on vacancy emission; sustained vacancy emission will undoubtedly exhaust its vacancy supply. Similarly, when serving as a sink, a GB also has an intrinsic upper limit for vacancy accommodation; prolonged vacancy absorption will eventually lead to saturation. This defines the **capacity** of GBs to act as a source or sink for vacancies. More importantly, in their role as sources or sinks, GBs themselves necessarily undergo structural changes upon absorbing or emitting vacancies. Specifically, as the vacancy flux increases, these structural changes become more pronounced, culminating in a state of exhaustion or saturation—a physical manifestation of the capacity. This capacity issue is also intimately intertwined with another long-standing open question: whether the vacancy concentration within individual grains of a polycrystalline material can attain thermal equilibrium. If GBs exhibit a finite vacancy sink/source capacity, the achievement of a uniform equilibrium vacancy concentration within grains would be hindered. Conversely, the sustained establishment of such uniform equilibrium vacancy concentration across grains would imply that the sink/source capacity of GBs is effectively unbounded—a scenario that logically runs counter to the fundamental definition of a “source and sink”. Yet, to the best of our knowledge, this capacity has never been systematically characterized. In fact, a comprehensive investigation into the dynamic mechanisms underpinning intensive vacancy generation and annihilation mediated by GBs remains notably lacking.

In this study, we present the first direct and comprehensive observation of the GB-mediated vacancy thermal equilibration (VTE) process in a face-centered cubic (fcc) Al system, in which a pair of low-angle tilt GBs is rationally engineered as the sole vacancy source and sink. By leveraging the time-averaged atomic volume spectroscopy (TAVS) method³⁶—a recently developed, state-of-the-art technique for vacancy identification—we monitored the real-time, concurrent evolution of GBs and vacancies across the entire VTE process. The principle underpinning TAVS centers on identifying the atomic “cages” that enwrap vacancies, rather than the vacancies themselves³⁶. Specifically, Voronoi tessellation analysis discriminates between vacancy-enwrapping atoms (VAs) and normal atoms (NAs) based on their time-averaged atomic volumes. Subsequent nearest-neighbor analysis clusters these VAs into discrete cages, thereby enabling comprehensive characterization of the intrinsic properties of individual vacancies—including their spatial location, size, volume, morphology, and energy state. It is precisely this key capability of TAVS—accurate real-time vacancy

identification—that facilitated the first direct observation of the full GB-mediated VTE process.

Results

Initial configurations

Figure 1a depicts the structure of an ideal fcc Al bicrystal, representing the original configuration, in which a $\Sigma 145(0\ 1\ 12)/[100]$ tilt GB forms between two grains with a tilt angle of 9.5° . Owing to the periodic boundary conditions, a pair of parallel tilt GBs is generated. Each tilt GB consists of six edge dislocations, with adjacent dislocations spaced $4.5a$ apart (where a denotes the lattice constant). The structural details of these GB dislocations are presented in Fig. 1b, c. Underneath each extra half-atomic plane, shield-shaped pentagons—each comprising 8 atoms, as visualized in the $[100]$ projection of Fig. 1b—delineate the critical **vacant channels** within the edge dislocations. As indicated by the circle and square symbols in Fig. 1b, the 8 atoms delineating each vacant channel are arranged in two distinct layers. These vacant channels are interconnected via the half-atomic planes of the edge dislocations (the black solid lines in Fig. 1b). Collectively, these extra half-atomic planes constitute the structural backbone of the tilt GB, which we term its “vertebral planes” (appearing as vertebral lines in the (100) projection of Fig. 1b). As the primary research candidate, this typical low-angle tilt GB features prominent structural stability and homogeneity, which allows us to eliminate extraneous confounding factors and focus exclusively on the structural evolution of GBs induced by vacancy generation and annihilation.

Previous studies have typically characterized GBs based on atomic energy, and we adopted the same approach herein, with the corresponding results presented in Fig. 1c. Concurrently, we employed the Voronoi atomic volume metric inherent to the TAVS method to characterize GBs, as this metric enables precise quantification of lattice distortion—a capability not afforded by atomic energy alone. As elaborated in subsequent sections, the lattice distortion regions along GBs serve as a key determinant in delineating the pathways governing vacancy emission and absorption by GBs. We therefore utilized the normalized atomic volume ($\Omega_A = V_A/V_0$) as the metric for characterizing distortion regions (as visualized in Fig. 1b), where V_A denotes the instantaneous volume of a given atom, and V_0 represents the average volume of atoms in a defect-free lattice (i.e., the standard atomic volume). Per the color coding scheme in Fig. 1: $\Omega_A > 1$ indicates that the local region is under tensile stress (colored yellow); $\Omega_A = 1$ corresponds to a distortion-free lattice region; $\Omega_A < 1$ signifies that the local region is under compressive stress (colored red). Furthermore, a larger magnitude of deviation of Ω_A from 1—reflected by the depth of the color—correlates with a more pronounced degree of local distortion and the associated stress level. Accordingly, we adopted a $\pm 1\%$ volume deviation threshold—corresponding to ~ 2.738 GPa and ~ 0.756 GPa—to demarcate compressive (upper red region) and tensile (lower yellow region) regions, as illustrated in Fig. 1b.

Building upon the original configuration, three initial samples with distinct vacancy concentrations were devised to facilitate the simulations of the VTE process: an undersaturation state (denoted as UE* in Fig. 1d) and two supersaturation states (denoted as SE in Fig. 1e and SE* in Fig. 1f, respectively). To be specific, UE* is vacancy-free, whereas SE and SE* contain 400 and 640 uniformly distributed monovacancies—corresponding to a supersaturated concentration of 1.28‰ and the maximum achievable limit of 2.04‰ (i.e., the solidification-trapped vacancy concentration³⁶), respectively. Blue atoms in Fig. 1e, f denote VA cages identified via the TAVS method. Subsequently, all samples were subjected to isothermal-isobaric relaxation at 900 K (note that the theoretical melting point of the adopted interatomic potential³⁷ is 926.15 K) to investigate massive vacancy absorption and emission mediated by the tilt GBs.

Visualization of the VTE process

VTE simulations were conducted under both the Gibbs (NPT: constant number of atoms, pressure, and temperature) and canonical (NVT: constant number of atoms, volume, and temperature) ensembles. Specifically,

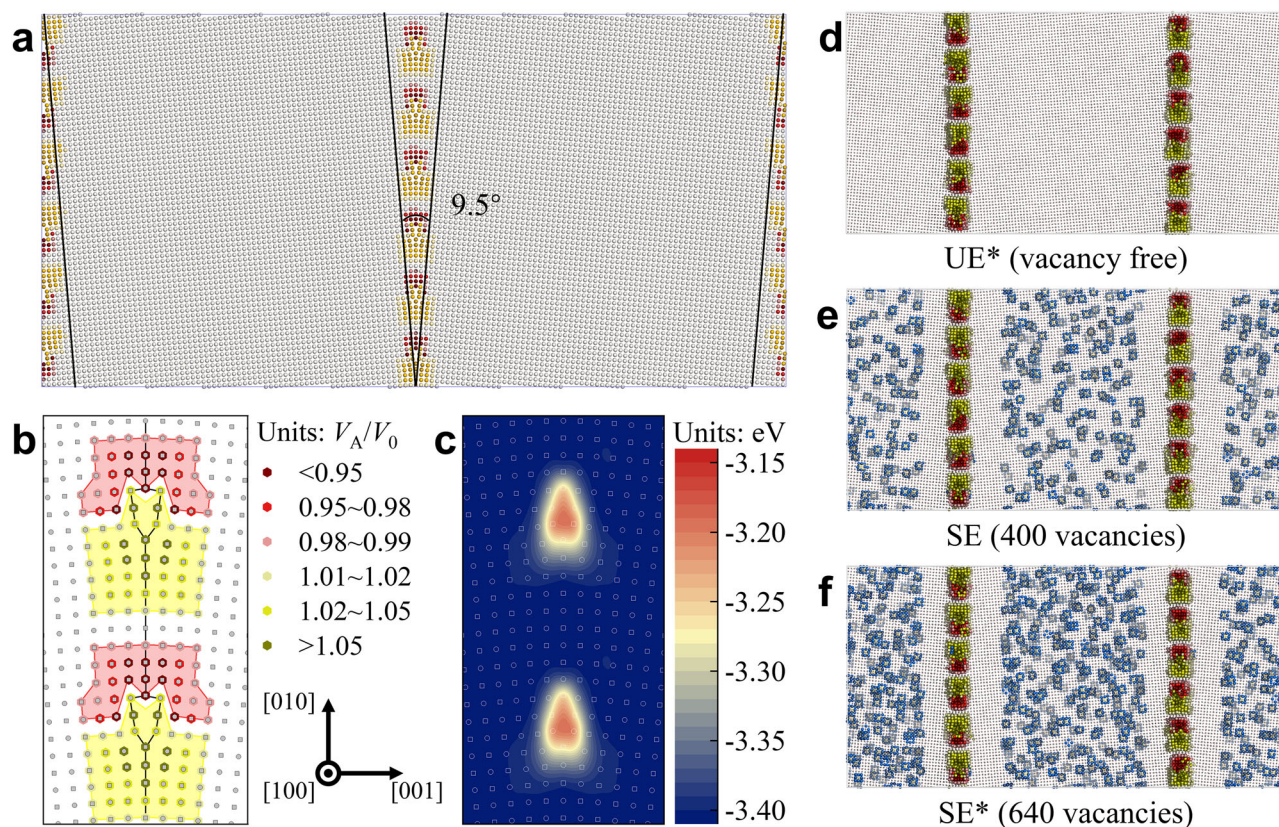


Fig. 1 | Original configuration of an ideal fcc-Al bicrystal containing a pair of $\Sigma 145(0\ 1\ 12)$ tilt GBs, together with initial configurations of UE*, SE, and SE* (viewed along the $[100]$ direction). a Structure of the ideal bicrystal with paired tilt GBs. **b** Detailed structure of the tilt GB: red and yellow represent compressive and

tensile regions, respectively. **c** Atomic energy map of the tilt GB. **d–f** Initial configurations at 900 K: vacancy-free UE*, SE with $C_V = 1.28\%$, and SE* with $C_V = 2.04\%$; VAs are highlighted in blue.

the Gibbs ensemble corresponds to an idealized scenario in which GBs are influenced solely by vacancy absorption and emission, thus remaining unperturbed by hydrostatic stress induced by massive vacancy generation and annihilation. In contrast, the canonical ensemble—characterized by a fixed volume—mimics the intrinsically constrained environment of grains in polycrystalline materials, such that the GBs experience the synergistic effect of both aforementioned processes. Figure 2 visualizes the VTE processes via representative snapshots of UE* and SE*, accompanied by magnified views highlighting the corresponding GB structures. For UE* (Fig. 2a: NPT; Fig. 2c: NVT), all vacancies were emitted exclusively by the tilt GBs and migrated toward the grain interiors. Conversely, all supersaturated vacancies in SE* were absorbed by the GBs to reduce the vacancy concentration (Fig. 2b: NPT; Fig. 2d: NVT). Ultimately, both UE* and SE* attained their respective equilibrium vacancy concentrations—albeit with minor discrepancies—and exhibited a uniform spatial distribution. Notably, all vacancies were generated or annihilated exclusively at the tilt GBs—none occurred within the grain interiors. These results confirm that tilt GBs function as dual vacancy sources and sinks: they emit or absorb vacancies depending on the saturation state of the grains, thereby fulfilling their role in mediating vacancy equilibration.

Another central focus of this study is the structural evolution of the tilt GBs during the VTE process. To visualize the structural evolution, we highlight GB distortion regions in red and yellow in the upper overview panels, and delineate the cross-sectional profiles with white lines in the lower magnified views—specifically outlining the eight-atom configuration of the vacant channels. The GB structure observed in Fig. 2 deviates slightly from the ideal configuration depicted in Fig. 1b, and it features a tree-like branched cross-sectional morphology (see the lower magnified views), with its vacant channels transitioning from shield-shaped pentagons to hexagons. Strikingly, this GB structure remained virtually intact throughout the entire

VTE process, irrespective of initial conditions (UE* vs. SE*) and thermodynamic ensembles (NPT vs. NVT). Neither the cross-sectional profiles nor the GB distortion regions—down to the positions of individual dislocations and their spacing—exhibited any discernible structural variation. This structural invariance of the tilt GBs suggests that vacancy absorption and emission do not elicit structural modifications in the GBs themselves. Given that this observation is at odds with previous reports^{15,17,21,34,38–40}, we conducted a stringent quantitative analysis of the VTE process to substantiate this unexpected structural invariance.

Quantification of the VTE process

Figure 3 comprehensively illustrates the real-time quantitative monitoring of the VTE process via four key physical parameters: vacancy concentration (C_V), energy, volume, and hydrostatic stress (σ). More specifically, the associated changes in energy and volume are further resolved into three distinct components: the GB energy and excess volume ($\Delta\gamma_{GB}$ and $\Delta V_{GB,ex}$), the total system energy and volume (ΔE_{Sys} and ΔV_{Sys}), and the total vacancy formation energy and volume (ΔE_{Vac} and ΔV_{Vac}). The definitions of the former two GB-related quantities are provided in the Methods section.

We first examined the temporal evolution of vacancy concentration (C_V)—the primary metric for characterizing the VTE process. Irrespective of the ensemble [NPT (Fig. 3a) or NVT (Fig. 3f)] or the sample type (UE*, SE, or SE*), C_V exhibits clear relaxation toward its equilibrium value, with all samples reaching equilibrium uniformly within 100 ns. The only notable difference lies in their convergence behavior: the three samples within the NPT ensemble converge to an identical equilibrium concentration, whereas their NVT counterparts exhibit minor discrepancies. Specifically, the equilibrium concentration achieved under the NPT ensemble is $(4.293 \pm 0.396) \times 10^{-4}$, consistent with our previously reported value under identical thermodynamic conditions⁴¹. In contrast, the three NVT samples

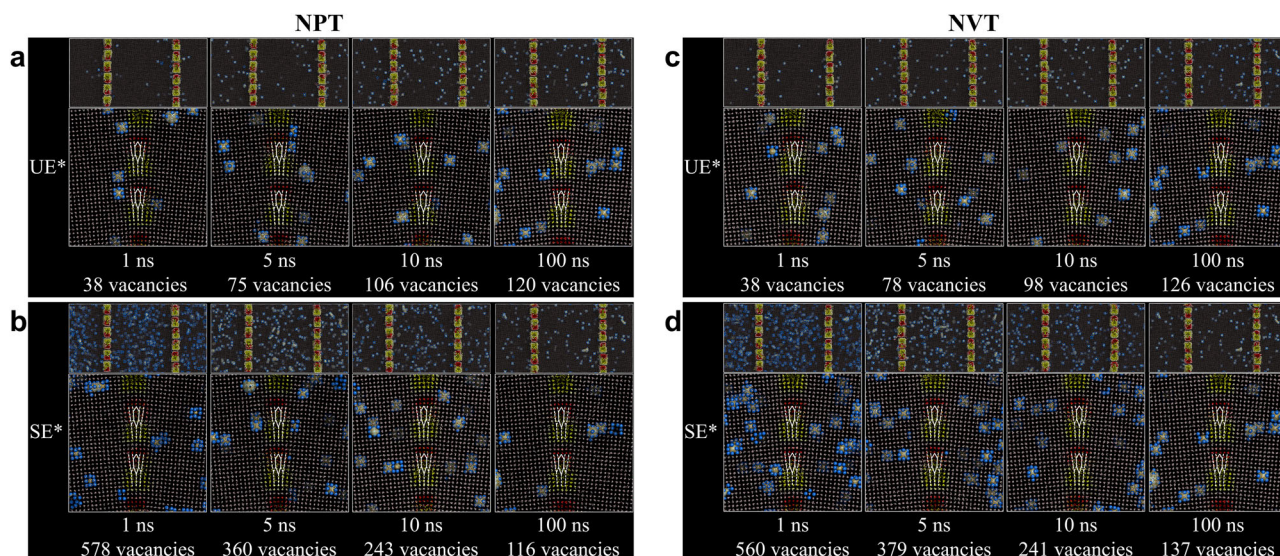


Fig. 2 | Visualization of the VTE process: a, b NPT ensemble, c, d NVT ensemble. Each experiment contains 4 snapshots. Each snapshot includes an overview (upper panel), where the GBs are delineated by distortion regions, and a magnified view (lower panel), with GB morphology outlined by white lines.

Fig. 3 | Temporal evolution of key physical parameters during the VTE process: a–e NPT ensemble, f–j NVT ensemble. a, f Vacancy concentration (C_V); **b, g** GB energy change ($\Delta\gamma_{GB}$); **c, h** System energy change (ΔE_{Sys}) and nominal change of vacancy formation energy (ΔE_{Vac}); **d, i** GB excess volume change ($\Delta V_{GB,ex}$), system volume change (ΔV_{Sys}), and nominal change of vacancy formation volume (ΔV_{Vac}); **e, j** hydrostatic stress (σ_{Sys}).

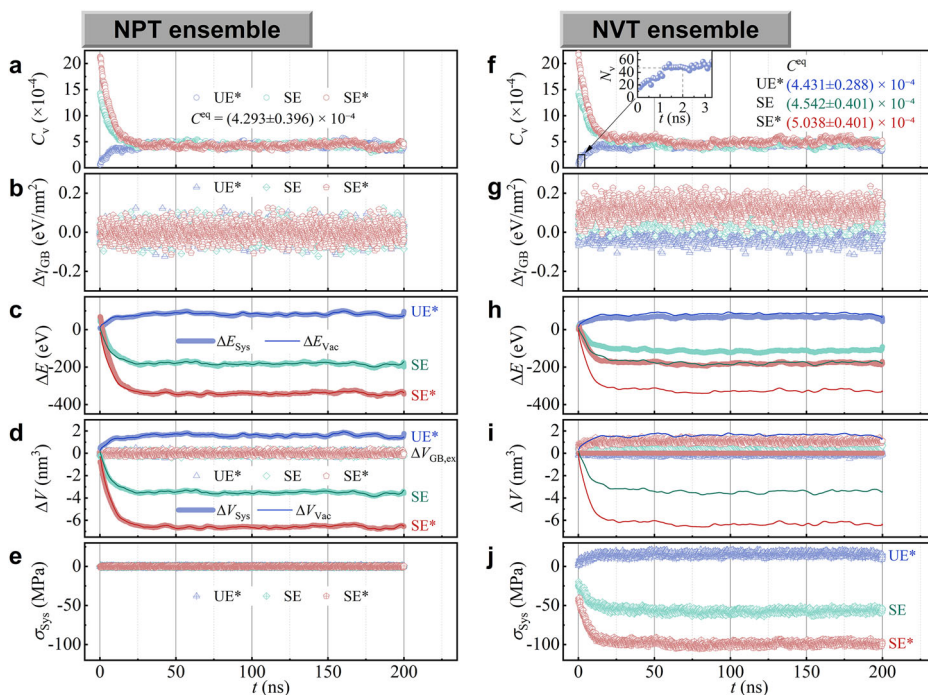


exhibit marginally divergent equilibrium concentrations. These subtle discrepancies stem from the volumetric constraint inherent to the NVT ensemble. Notably, the substantial volume changes associated with extensive vacancy generation or annihilation are constrained under NVT conditions, which in turn induces hydrostatic stress within the system. This induced stress perturbs the system's thermodynamic equilibrium, thereby modulating the equilibrium vacancy concentration. Such behavior is expected to represent a pervasive scenario in polycrystalline systems.

Building on the findings from visualizing the VTE process—where all vacancy generation and annihilation events were found to occur at GBs—we further monitored the temporal evolution of energy and volume for the system, vacancies, and GBs (Fig. 3b–e, g–j). Our aim is twofold: to characterize the evolution of GBs, and to examine how GB variations and vacancy changes collectively govern the overall evolution of the entire

system. For the NPT ensemble, Fig. 3b depicts the temporal evolution of the GB energy change ($\Delta\gamma_{GB}(t) = \gamma_{GB}(t) - \gamma_{GB}(0)$) for the three samples. The initial value, $\gamma_{GB}(0) = 2.843 \text{ eV/nm}^2$, is in good agreement with the model prediction reported by Bulatov et al.⁴². Similarly, Fig. 3d illustrates the temporal evolution of the GB excess volume change ($\Delta V_{GB,ex}(t) = V_{GB,ex}(t) - V_{GB,ex}(0)$), where $V_{GB,ex}(0) = 9.261 \text{ nm}^3$. Irrespective of the initial vacancy concentrations, both $\Delta\gamma_{GB}(t)$ and $\Delta V_{GB,ex}(t)$ values fluctuate around zero over the entire VTE process. This observation indicates that the GBs in the NPT ensemble remain *unexpectedly invariant* during the entire VTE process, in terms of both their energy and volume. In contrast, the system energy change ($\Delta E_{Sys}(t) = E_{Sys}(t) - E_{Sys}(0)$) plotted in Fig. 3c and the system volume change ($\Delta V_{Sys}(t) = V_{Sys}(t) - V_{Sys}(0)$) shown in Fig. 3d exhibit pronounced variations, as indicated by the thick lines. For comparison, we computed the nominal changes in vacancy formation energy and volume,

Table 1 | The equilibrium values of key physical quantities: vacancy concentration (C^{eq}), GB energy and GB excess volume change (γ_{GB} , $\Delta V_{\text{GB,ex}}$), total system energy and volume changes (ΔE_{Sys} , ΔV_{Sys}), nominal changes of vacancy formation energy and volume (ΔE_{Vac} , ΔV_{Vac}), hydrostatic stress (σ_{Sys})

	NPT	NVT
C^{eq} ($\times 10^{-4}$)	4.293 $\pm$ 0.396	4.431 $\pm$ 0.288 (UE*)
		4.542 $\pm$ 0.401 (SE)
		5.038 $\pm$ 0.401 (SE*)
γ_{GB} (eV·nm ²)	2.843 $\pm$ 0.036	2.838 $\pm$ 0.035 (UE*)
		2.924 $\pm$ 0.036 (SE)
		2.963 $\pm$ 0.035 (SE*)
ΔE_{Sys} (eV)	81.93 $\pm$ 7.908 (UE*) −183.2 $\pm$ 20.72 (SE) −338.7 $\pm$ 20.47 (SE*)	66.14 $\pm$ 17.31 (UE*) −113.4 $\pm$ 17.43 (SE) −179.2 $\pm$ 17.17 (SE*)
ΔE_{Vac} (eV)	82.56 $\pm$ 8.038 (UE*) −182.4 $\pm$ 7.189 (SE) −340.2 $\pm$ 7.597 (SE*)	85.11 $\pm$ 5.547 (UE*) −177.0 $\pm$ 7.674 (SE) −325.8 $\pm$ 7.689 (SE*)
$\Delta V_{\text{GB,ex}}$ (nm ³)	9.261 $\pm$ 0.123	9.193 $\pm$ 0.113 (UE*) 9.963 $\pm$ 0.119 (SE) 10.341 $\pm$ 0.120 (SE*)
ΔV_{Sys} (nm ³)	1.588 $\pm$ 0.214 (UE*) −3.539 $\pm$ 0.202 (SE) −6.572 $\pm$ 0.203 (SE*)	0
ΔV_{Vac} (nm ³)	1.599 $\pm$ 0.156 (UE*) −3.532 $\pm$ 0.139 (SE) −6.589 $\pm$ 0.147 (SE*)	1.649 $\pm$ 0.107 (UE*) −3.428 $\pm$ 0.149 (SE) −6.311 $\pm$ 0.149 (SE*)
σ_{Sys} (MPa)	0	15.83 $\pm$ 2.20 (UE*) −57.05 $\pm$ 2.34 (SE) −98.51 $\pm$ 2.31 (SE*)

defined as $\Delta E_{\text{Vac}}(t) = [N_{\text{Vac}}(t) - N_{\text{Vac}}(0)]\Delta e_f$ and $\Delta V_{\text{Vac}}(t) = [N_{\text{Vac}}(t) - N_{\text{Vac}}(0)]\Delta v_f$, respectively, based exclusively on the change in vacancy count. Here, $N_{\text{Vac}}(t)$ and $N_{\text{Vac}}(0)$ denote instantaneous and initial vacancy numbers, respectively, while $\Delta e_f = 0.66$ eV and $\Delta v_f = 0.01278$ nm³ correspond to the standard formation energy and volume of a monovacancy in fcc-Al at 900 K and 0 bar⁴³. These nominal changes are plotted as thin lines in Fig. 3c, d for direct comparison with the system quantities represented by the thick lines. Strikingly, *the variations in system energy and volume changes are in perfect agreement with those of the nominal vacancy formation energy and volume throughout the entire VTE process*. This perfect agreement, along with the invariance of GBs, unequivocally demonstrates that the observed changes in system energy and volume are attributed exclusively to vacancy dynamics, with no contribution whatsoever from the GBs. These observations quantitatively corroborate the structural invariance of GBs, as previously identified via visualization of the VTE process.

We now shift our focus to the NVT ensemble. As emphasized previously, the NVT ensemble mimics the constant-volume constraint inherent to polycrystalline systems. During intensive vacancy generation and annihilation (i.e., the VTE process), the NPT ensemble sustains a zero-pressure state by adjusting the system volume—thereby decoupling the effects of vacancy activity from those of the induced hydrostatic stress. In contrast, the NVT ensemble develops a non-zero hydrostatic stress as a direct consequence of volume fixation, which in turn modulates the energy and volume of both GBs and grains. As illustrated by the evolution of hydrostatic stress σ_{Sys} in Fig. 3e, j, stress remains strictly zero in the NPT ensemble; in the NVT ensemble, however, stress rises with vacancy accumulation in UE* and declines with vacancy depletion in SE and SE*.

Ultimately, the equilibrium hydrostatic stresses for the three NVT samples are determined to be 15.83 $\pm$ 2.20 MPa, −57.05 $\pm$ 2.34 MPa, and −98.51 $\pm$ 2.31 MPa, respectively. This buildup of hydrostatic stress modulates the overall VTE process. First, as depicted in Fig. 3f, the final equilibrium concentrations exhibit slight disparities across the three samples. Second, GBs and grains undergo densification or intra-structural dilation to accommodate the net change in vacancy volume, as illustrated in Fig. 3i. Accordingly, the GBs in UE* are subjected to a compressive stress state, whereas those in SE and SE* experience tensile stress. Third, the variation in NVT system energy therefore stems not only from the vacancy generation and annihilation, but also from such densification and dilation processes. Thus, as shown in Fig. 3h, the changes in system energy and vacancy formation energy no longer align; instead, the magnitude of this discrepancy scales with the amplitude of hydrostatic stress. Finally, this effect directly induces subtle variations in GB energy, which, as illustrated in Fig. 3g, deviate from the zero baseline.

Nonetheless, we highlight that even the minor changes in GB structure and energy within the NVT ensemble bear no direct relation to vacancy generation and annihilation. Collectively, our results demonstrate that vacancy generation and annihilation do not directly induce structural variations in GBs, regardless of whether the system is governed by the NPT or NVT ensemble. Incidentally, the average values of all parameters during the equilibrium stage (100–200 ns), as presented in Fig. 3, are summarized in Table 1 as equilibrium reference data.

Based on the preceding visualization and quantitative monitoring of the VTE process, we draw two critical conclusions.

- (1) Our results unambiguously demonstrate that tilt GBs absorb and emit vacancies, thereby functioning as both vacancy sources and sinks to mediate the VTE process. This conclusion is consistent with the current understanding established from extensive experimental observations and theoretical descriptions.
- (2) Notably, our findings reveal unequivocally that the vacancy absorption or emission exerts no discernible influence on the grain boundaries themselves. Specifically, throughout the VTE process, despite the extensive vacancy generation and annihilation, GBs remain structurally invariant. Even the subtle variations observed in the NVT ensemble are attributable to the induced hydrostatic stress, rather than to vacancy absorption or emission events themselves. This finding challenges the prevailing paradigm and provides a fundamentally distinct perspective on the role of GBs in mediating vacancy generation and annihilation.

Collectively, these two conclusions establish a novel conceptual framework for the GB-vacancy interaction mechanism: tilt GBs act as dual **catalysts** for vacancy generation and annihilation, thereby facilitating the VTE process. On the one hand, this mechanism can be rationalized in terms of thermodynamic driving forces and kinetic energy barriers. As vacancies are thermodynamic equilibrium defects, any deviation of the intragranular vacancy concentration from its equilibrium value—whether undersaturation or oversaturation—creates a thermodynamic driving force that drives the system toward equilibrium. However, the energy barrier associated with direct vacancy generation or annihilation within grain interiors is prohibitively high, rendering this pathway kinetically inaccessible. In contrast, this barrier is drastically reduced at tilt GBs^{44–46}. In other words, the catalytic role of tilt GBs stems from their propensity to drastically lower the energy barrier for the vacancy generation and annihilation. On the other hand, a hallmark feature of a catalyst is that it retains its chemical identity and structural integrity unchanged following the catalytic process. Notably, the structural and energetic invariance of tilt GBs throughout the VTE process aligns perfectly with this canonical definition of catalysis. On this basis, we put forward a catalytic mechanism to clarify the fundamental nature of GB-vacancy interactions.

The foregoing comprehensive quantitative monitoring of the VTE process provides robust corroboration for the proposed catalytic mechanism. Nevertheless, two key microscopic details remain to be fully elucidated.

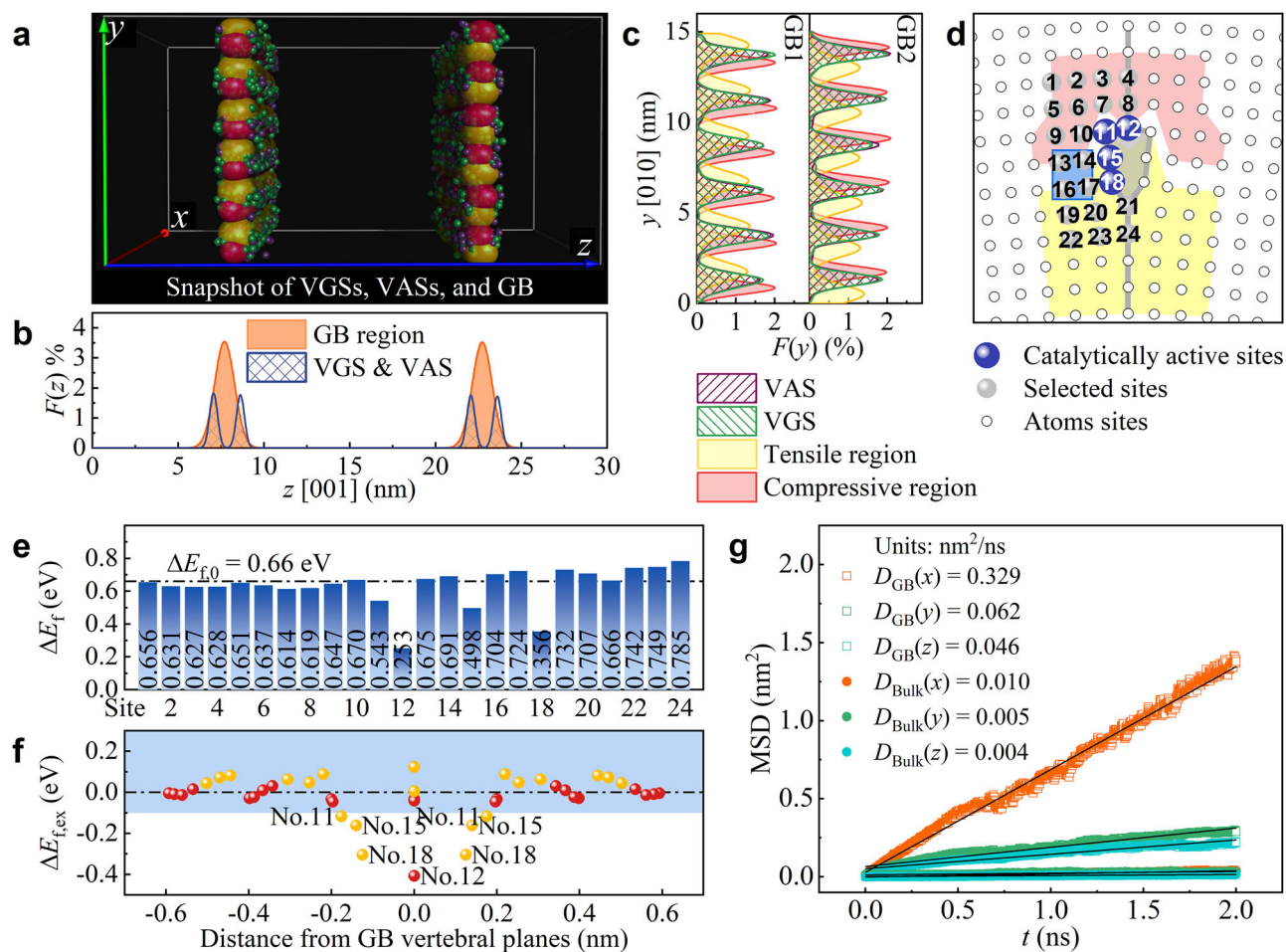


Fig. 4 | Schematic illustration of catalytically active sites and GBs' structural invariance. **a** Spatial distribution of vacancy generation sites (VGSs) and vacancy annihilation sites (VASs); **b** Projections of VGS/VAS and GB distributions on the z -axis; **c** Projection of VGS/VAS and GB distributions on the y -axis; **d** Selected lattice sites surrounding vacant channels; **e** Vacancy formation energy (ΔE_f) at selected

lattice sites within GBs; **f** Deviation of ΔE_f from the standard value $\Delta E_{f,0} = 0.66$ eV as a function of distance from the GB vertebral plane; **g** 1D mean squared displacements (MSDs) of GBs and grain interiors, together with the corresponding diffusion coefficients.

First, the specific atomic mechanisms underpinning vacancy absorption and emission at tilt GBs—particularly the nature of the catalytically active sites—require explicit clarification. Second, the structural invariance of tilt GBs following these vacancy exchange events calls for in-depth unraveling. Addressing these two critical questions will facilitate the delineation of a complete and consistent catalytic cycle.

Catalytically active sites, GBs' invariance and catalytic cycle

To elucidate the catalytically active sites and the structural invariance of GBs, we conducted a high-resolution temporal sampling analysis of the initial 2 ns of the VTE process under the NVT ensemble, as illustrated in the inset of Fig. 3f. The objective herein is to pinpoint all vacancy generation sites (VGS) and vacancy annihilation sites (VAS). The methodology for identifying VGSs and VASs is described in detail in the Methods section. In a prior study³⁶, we quantified the jumping frequency of monovacancies in fcc-Al at 900 K as 128.75 ns^{-1} . This translates to a mean residence time of 7.77 ps between consecutive vacancy jumps. Accordingly, the average minimum duration for a round-trip vacancy jump event is 15.54 ps, which corresponds precisely to the average minimum lifetime of the vacancy generation/annihilation round-trip. Based on this data, the time step employed for VGS/VAS analysis was set to 1 ps. This time step is considerably shorter than the average minimum lifetime of 15.54 ps, thereby providing sufficiently high temporal resolution to ensure that essentially no vacancy generation or annihilation events are missed. Leveraging the 1 ps time step

—which yields a total of 2×10^3 configurations—every vacancy event occurring over the 2 ns duration was fully documented, with their respective VGSs and VASs identified. In total, 1303 VGSs and 1256 VASs were pinpointed, resulting in a net gain of 47 vacancies over this 2 ns equilibration period. In Fig. 4a, we visualize all VGSs (green spheres) and VASs (purple spheres), together with the GB distortion region model—where semi-transparent red and yellow isosurfaces delineate the compressive and tensile regions, respectively. Meanwhile, we projected this three-dimensional distribution onto the z - and y -axes to generate one-dimensional (1D) distribution profiles, as presented in Fig. 4b, c, respectively. We further overlaid the GB regions to clarify the spatial relationship between VGSs/VASs and GBs. Specifically, Fig. 4c plots the distribution profiles for the compressive and tensile regions within GBs, respectively.

We first examine the spatial distribution of VGSs and VASs. Figure 4b presents a lateral view of the tilt GBs, demonstrating that all VGSs and VASs are located in immediate proximity to both sides of the tilt GBs. Furthermore, Fig. 4c—which presents the projection along the direction perpendicular to the GB vertebral plane—illustrates that, on the one hand, the distributions of VGSs and VASs are spatially coincident; on the other hand, these sites are localized at the interfaces between the compressive and tensile regions of the dislocations, precisely corresponding to the vacant channels. Synthesizing these observations, we postulate that all vacancy generation and annihilation events nucleate within these vacancy channels. Consequently, the catalytically active sites of GBs—responsible for nucleating

vacancy generation and annihilation—are spatially correlated with these vacant channels. Another notable observation is that all VGSs and VASs correspond exclusively to monovacancy nucleation sites. This implies that only monovacancies can be directly generated and annihilated at tilt GBs, whereas divacancies and larger vacancy clusters cannot undergo direct generation or annihilation via this pathway.

Counts of the 1303 VGSs and 1256 VASs demonstrate that, over the 2 ns equilibration period, the tilt GBs within the system emitted 1303 vacancies and absorbed 1256 vacancies in total. Assuming that only 47 vacancies migrated successfully into the grain interiors, this implies that 1256 vacancies were generated and subsequently annihilated instantaneously at the GBs. Given that the system comprises two tilt GBs—each hosting six dislocation lines with a length of ~ 12.138 nm—the corresponding vacancy generation/annihilation rate is $8.62 \text{ ns}^{-1} \cdot \text{nm}^{-1}$. Since the high temporal resolution of 1 ps employed in the VGS/VAS analysis ensures the comprehensive sampling of all vacancy events, this rate may be considered the definitive vacancy generation/annihilation rate for edge dislocations in fcc-Al at 900 K. Such a high rate demonstrates that not only are the energy barriers for initial vacancy generation and annihilation within vacant channels significantly lower, but also atomic migration within these channels proceeds at an exceptionally high rate. This behavior stems from the fact that any vacancy generation or annihilation event inherently corresponds to the formation or removal of a surplus atom, which must undergo rapid migration within the channel to sustain the high vacancy generation/annihilation rate across the entire channel. This mechanistic understanding is critical, as it fully accounts for the structural invariance of GBs during dynamic vacancy generation and annihilation processes.

The distribution of VGSs and VASs immediately flanking both sides of the vacant channels reveals that the catalytically active sites directly correspond to lattice sites surrounding these channels. To further pinpoint these sites, we selected 24 lattice sites adjacent to the vacant channel—as depicted in Fig. 4d—and computed the vacancy formation energy (ΔE_f) at each site. These computations were carried out using the molecular mechanics method for the original configuration presented in Fig. 1, employing the same interatomic potential used in the MD simulations. Figure 4e displays the complete set of ΔE_f values, with the standard vacancy formation energy ($\Delta E_f = 0.66$ eV) indicated by a horizontal dashed line. In Fig. 4f, we further plot the deviation of the computed vacancy formation energy from the standard value—defined as $\Delta E_{f,\text{ex}} = \Delta E_f - \Delta E_{f,0}$ —as a function of the distance from the vertebral plane of the GB. Furthermore, we map the peak positions and full-width at half-maximum values extracted from the VGS and VAS distribution profiles in Fig. 4b, c onto the blue box highlighted in Fig. 4d. We find that this box coincides exactly with lattice sites 13, 14, 16, and 17. These four sites therefore constitute the predominant loci for VGS and VAS events.

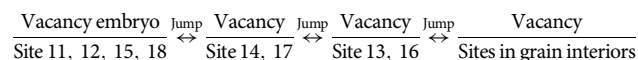
The vacancy formation energy profile shown in Fig. 4e, illustrates that, with the exception of sites 11, 12, 15, and 18, ΔE_f values at all other sites virtually fluctuate around the standard value ($\Delta E_{f,0} = 0.66$ eV). Specifically, ΔE_f values in the compressive region are slightly lower than or equal to $\Delta E_{f,0}$, whereas those in the tensile region are marginally higher. In contrast, the ΔE_f values at the four exceptional sites—corresponding to 7 sites within the 8-site ring surrounding the vacancy channel—are markedly lower than $\Delta E_{f,0}$, with reductions of 17.7%, 24.5%, 46%, and 61.7% for sites 11, 15, 18, and 12, respectively, relative to the 0.66 eV standard. This marked reduction in vacancy formation energy at these four exceptional sites provides compelling evidence for the ultra-high vacancy generation/annihilation rate inferred from our earlier statistical analyses of VGS and VAS populations, and thereby confirms that vacancy generation or annihilation is considerably facilitated therein. In other words, *these four sites constitute the definitive catalytically active sites* for vacancy generation and annihilation.

With these results, we can now establish the complete catalytic mechanism at the atomic scale.

(1) Vacancy generation and annihilation events initiate at the catalytically active sites—specifically, the seven sites within the 8-site ring surrounding the vacancy channel, namely sites 11, 12, 15, and 18 in

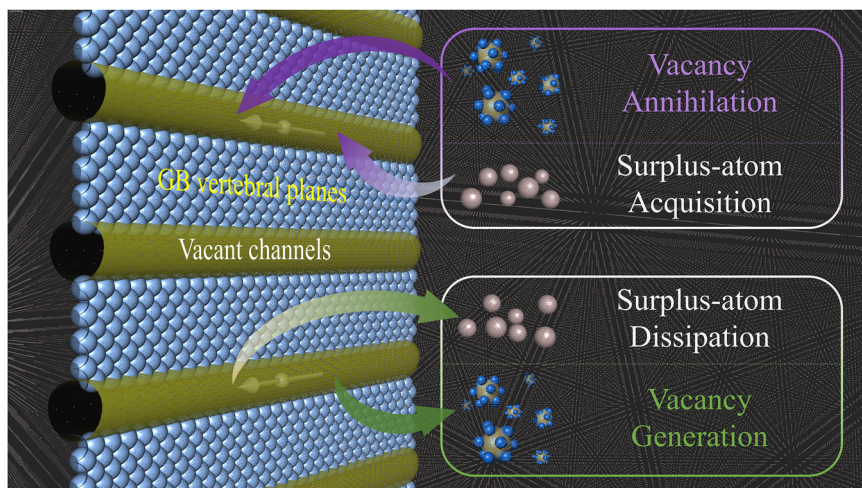
Fig. 4d. The vacancies formed at these sites can be defined as **vacancy embryos**, because the atomic cages enwrapping these vacancies exhibit defective rather than ideal configurations (e.g., they lack the characteristic 12-atom coordination environment).

(2) Sites 14 and 17 are identified as the genuine VGSs and VASs. Although the blue box in Fig. 4d also labels sites 13 and 16 as VGS/VAS loci, these two sites are not authentic, as they are not adjacent to the catalytically active sites. Rather, vacancies migrate between sites 14/17 and sites 13/16 within the 1 ps time interval of VGS/VAS determination, leading to their misidentification as VGSs/VASs. Importantly, the frequent misidentification of sites 13 and 16 as VGSs/VASs reveals that vacancies migrate along the $\{14, 17\} \leftrightarrow \{13, 16\}$ pathway, confirming that the compressive-tensile region interface is precisely this migration pathway. Notably, site 14 is adjacent to three catalytically active sites (i.e., sites 11, 15, and 18), whereas site 17 is adjacent to sites 15 and 18. In particular, site 12—identified as the most catalytically potent site—exhibits the highest probability of vacancy embryo formation, due to its vacancy formation energy being reduced by up to 61.7%. However, the vacancy embryos formed at this site rarely diffuse outward directly; instead, they must first migrate through site 11, 15, or 18 before migrating to sites 14 and 17, where they transform into genuine vacancies. Accordingly, a well-defined vacancy generation/annihilation and migration pathway between GBs and grain interiors can be delineated as follows:



(3) Although the vacancy generation/annihilation pathway has been delineated, one final key issue remains to be addressed. When a vacancy embryo is generated or annihilated within the vacant channel, this process is inevitably accompanied by the concomitant generation or consumption of a surplus atom within the channel: the formation of a vacancy embryo coincides with the ejection of a surplus atom, whereas the annihilation of a vacancy embryo requires the incorporation of such a surplus atom. Historically, the prevailing view regarding this phenomenon has maintained that GBs accommodate such surplus atoms via internal structural relaxation, a process that would inevitably alter both the structure and energetics of GBs. However, our results, quantitatively validated in Fig. 3, clearly establish that GBs do not accommodate these surplus atoms internally at all. Instead, GBs must dissipate (or acquire) such surplus atoms to (or from) adjacent grains owing to the robust structural invariance of GBs. Thus, the key unresolved issue pertaining to this final aspect is as follows: How does the GB dissipate (or acquire) such surplus atoms to (or from) the grains? The conventional view holds that vacancy generation/annihilation and the surplus atom dissipation/acquisition must occur as paired, spatially and temporally coupled events. Typically, within the grain interior, these two events coincide spatially and temporally: when an atom transforms into a surplus atom (self-interstitial), its original lattice site is concomitantly converted into a vacancy, and vice versa. In contrast, within the vacancy channels at GBs, the spatial and temporal decoupling of these paired events is feasible. It is precisely this decoupling that enables the absorption or emission of vacancies while maintaining the structural invariance of GBs. This spatiotemporal decoupling is substantiated by two key kinetic characteristics of the vacant channels. First, there is an ultrahigh vacancy generation/annihilation rate of $8.62 \text{ ns}^{-1} \cdot \text{nm}^{-1}$, a value that is in principle equivalent to the rate of surplus atom dissipation/acquisition. Such an ultrahigh rate underscores that vacant channels are extremely facile for the generation and annihilation of both vacancies and surplus atoms. Second, the vacant channels feature a critical prerequisite: the nearly unrestricted migration of surplus atoms within the channels. To validate this unrestricted atomic mobility, we partitioned the system into two distinct domains—grains and GBs—and computed the one-

Fig. 5 | Schematic catalytic circle of GB-mediated vacancy generation and annihilation at the atomic scale.



dimensional mean squared displacements (1D MSDs) along each of the three orthogonal axes. The corresponding results are presented in Fig. 4g, together with the corresponding diffusion coefficients. As observed, the atomic diffusion coefficient along the x -axis (aligned with the channel direction) within GBs is one order of magnitude greater than those along the other two axes within the GBs, and two orders of magnitude higher than those along all three axes within the grains. Further analysis reveals that atoms surrounding vacant channels within GBs account for only ~10 at.%. Assuming negligible diffusion of all other atoms in the GB domain, i.e., that the exceptionally high diffusion coefficient in Fig. 4g arises exclusively from these 10% of atoms surrounding vacant channels, this implies that the actual diffusion coefficient of atoms in vacant channels is another order of magnitude higher, reaching a value more than three orders of magnitude higher than that of the bulk. This finding completes the mechanistic framework of the present study, conclusively demonstrating that atoms diffuse extremely rapidly—i.e., nearly unrestrictedly—within the vacancy channels. In summary, it is precisely such an ultrahigh rate of vacancy generation/annihilation and surplus-atom dissipation/acquisition at the catalytically active sites—together with the ultrafast atomic mobility within the vacancy channels—that decouples vacancy events from surplus-atom events, rendering them relatively independent processes that can occur autonomously and be spatially and temporally separated. In turn, it is precisely this decoupling behavior of GBs that endows them with catalytic activity toward vacancy generation and annihilation.

This constitutes the complete catalytic mechanism governing vacancy generation and annihilation at grain boundaries, as illustrated in the schematic of GB-catalyzed vacancy generation/annihilation in Fig. 5. For the critical fundamental question raised at the outset of this study, the present mechanism furnishes a definitive resolution. In essence, this GB-mediated vacancy generation/annihilation process is intrinsically catalytic. As a direct corollary, tilt GBs exhibit an effectively unbounded capacity for vacancy absorption and emission.

Finally, imagine the scenario schematized in Fig. 5. Dislocation lines within low-angle tilt GBs continuously and rapidly generate and annihilate vacancy embryos, corresponding to the dissipation and acquisition of surplus atoms. When the vacancy concentration inside grains is at thermodynamic equilibrium, the rates of vacancy generation and annihilation at GBs are balanced. Conversely, any deviation from the equilibrium vacancy concentration disrupts the diffusion balance between intragranular vacancies and catalytically active sites, which in turn induces an imbalance between the vacancy generation and annihilation rates at GBs, leading to the net generation or annihilation of vacancies. Furthermore, the resulting

surplus atoms are instantaneously dissipated into or acquired from the adjacent grains via the ultrahigh surplus-atom dissipation/acquisition rates and unrestricted atomic mobility within vacant channels, thereby rapidly restoring the structural stability and integrity of low-angle tilt GBs. This completes the catalytic circle underlying low-angle tilt GB-mediated vacancy generation and annihilation.

Discussion

In this study, using the TAVS method for precise, real-time identification of vacancies, we have for the first time achieved molecular dynamics simulations of the VTE process, wherein low-angle tilt GBs serve as the exclusive vacancy source or sink. Both visualization and quantitative monitoring of the VTE processes conclusively reveal the following robust findings. (1) During the VTE process, GBs indeed function as dual vacancy sources and sinks—that is, all vacancy generation and annihilation events during the equilibration process originate at GBs, consistent with the conventional “source/sink” paradigm. (2) In contrast to this established paradigm, however, GBs preserve their structural integrity throughout the entire intensive vacancy generation/annihilation process. These observations lead us to propose a catalytic mechanism governing low-angle tilt GB-mediated vacancy generation and annihilation. To elucidate this catalytic mechanism at the atomic scale, we quantified the distribution of VGSs and VASs, calculated the vacancy formation energy at lattice sites surrounding vacant channels, and determined the one-dimensional mean squared displacement in both GBs and grain interiors. Based on these comprehensive results, we present an atomic-scale description of the catalytic mechanism governing low-angle tilt GB-mediated vacancy generation and annihilation. (1) Lattice sites surrounding the vacant channels exhibit extremely low vacancy formation energy, which markedly reduces the energy barrier for vacancy formation, thereby enabling high-density, high-rate vacancy embryo generation and annihilation. These sites are thus defined as catalytically active sites, satisfying the first prerequisite for a catalytic mechanism. (2) The ultrahigh atomic mobility within vacant channels allows vacancies to be generated or annihilated at specific locations, accompanied by the dissipation or acquisition of surplus atoms at distal sites within the channels. This unique decoupling behavior enables low-angle tilt GBs to preserve their structural invariance, fulfilling the second prerequisite for a catalytic mechanism. (3) Vacancy embryos continuously generated or annihilated at seven catalytically active sites within the eight-site ring surrounding each vacant channel migrate between GBs and adjacent grains via the interface between compressive and tensile regions, which acts as the critical migration pathway. Ultimately, the proposed catalytic mechanism provides a definitive answer not only to the “by what mechanism” component of the fundamental question posed at the outset, but also to the “to what extent” component: the capacity of GBs to act as vacancy sources or sinks is

effectively unbounded. The core physical basis of this catalytic mechanism originates from catalytically active sites inherent to vacant channels, coupled with ultrafast atomic migration that induces spatiotemporal decoupling of the inherently coupled vacancy and surplus atom events. We therefore hypothesize that all crystallographic defects featuring vacant channels—including high-angle tilt grain boundaries, pure edge dislocations, and mixed edge-screw dislocations—can exhibit catalytic activity toward vacancy generation and annihilation. Verifying this conjecture will form the central focus of our subsequent research efforts.

Methods

MD simulations of the VTE process

A bicrystal model containing 313,200 atoms was constructed to simulate the VTE process. The model comprises two fcc grains with a misorientation angle of 9.5° , each with dimensions of $12.138 \times 14.616 \times 14.616 \text{ nm}^3$. The system was then isothermally relaxed at 900 K for 200 ns to investigate the VTE behavior, with data output every 0.1 ns and an optimal averaging duration of 4 ps, as validated in our previous work³⁶. For the NPT ensemble, the system pressure was maintained at 0 bar; for the NVT ensemble, the system volume was fixed at the equilibrium value corresponding to 900 K and 0 bar. The equations of motion were integrated numerically using the Verlet-Leapfrog algorithm, with a time step of 2 fs. Temperature and pressure were controlled using the Nosé-Hoover thermostat and barostat⁴⁷, respectively. All simulations were performed using the LAMMPS package^{48,49}, and interatomic interactions of aluminum were modeled using the embedded-atom method (EAM) potential developed by Mendelev et al.³⁷.

GB delineation, GB energy and excess volume

GBs were delineated based on time-averaged atomic volume, consistent with the vacancy identification protocol employed in this study. Candidate GB atoms were screened using a $\pm 1\%$ volume deviation threshold relative to the atomic volume of defect-free sites.

The GB energy γ_{GB} is defined as the excess potential energy per unit GB area, expressed mathematically as:

$$\gamma_{\text{GB}} = \frac{1}{A_{\text{GB}}} \sum (e_{\text{GB}} - e_{\text{Coh}}), \quad (1)$$

where $A_{\text{GB}} \approx 370.688 \text{ nm}^2$ is the GB area, e_{GB} is the potential energy of individual GB atoms, and $e_{\text{Coh}} = -3.269 \text{ eV}$ represents the cohesive energy of defect-free lattice atoms measured at 900 K.

The GB excess volume $\Delta V_{\text{GB,ex}}$ is defined as the total excess volume of GBs relative to that of defect-free lattice atoms:

$$\Delta V_{\text{GB,ex}} = \sum (V_{\text{GB}} - V_0), \quad (2)$$

where V_{GB} is the volume of individual GB atoms, and $V_0 = 0.0177 \text{ nm}^3$ is the volume of defect-free lattice atoms measured at 900 K.

Vacancy formation energy

The vacancy formation energy (ΔE_f) at specific lattice sites surrounding the vacant channels of GBs was calculated using classical molecular mechanics within the LAMMPS package^{48,49}. Consistent with the MD simulations, the EAM potential developed by Mendelev et al.³⁷ was employed. Prior to ΔE_f calculations, the entire system containing the ideal tilt GBs was fully relaxed via the conjugate gradient algorithm, with convergence criteria of $10^{-5} \text{ eV}\cdot\text{nm}^{-1}$ per atom (force tolerance) and 10^{-6} eV (energy tolerance). Twenty-four representative lattice sites were then selected for calculation.

The vacancy formation energy ΔE_f at each selected site is defined as the energy difference between the supercell containing a single targeted monovacancy and the defect-free reference supercell, expressed

mathematically as:

$$\Delta E_f = E_{1v} - (E_{0v} - e_0), \quad (3)$$

where E_{1v} denotes the total energy of the supercell with one atom removed from the target site, E_{0v} is the total energy of the defect-free supercell, and $e_0 = -3.41 \text{ eV}$ is the energy of an isolated atom.

Identification of VGS and VAS

The criteria for identifying VGS and VAS are defined as follows. Based on the TAVS framework, a vacancy is identified by the atomic cage enveloping it. For any given vacancy, the relationship between its atomic cages at two consecutive time steps falls into one of the following categories:

- (1) Complete overlap: The atomic cages are identical, indicating that the vacancy remains stationary.
- (2) Partial overlap: For a monovacancy, this is characterized by exactly 5 overlapping atoms, corresponding to a single vacancy jump.
- (3) No overlap: This signifies a vacancy generation or annihilation event.

- If the vacancy cage exists only at the earlier time step but vanishes at the later step (i.e., all atoms constituting the cage revert from VA to NA), the event is classified as an annihilation.
- Conversely, if the vacancy cage appears only at the later time step, the event is classified as a generation.

The lattice site associated with such a generation or annihilation event is subsequently defined as a VGS or VAS, respectively, and recorded accordingly.

Data availability

All data supporting the findings of this study are available within the paper.

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Author contributions

Hao Wang performed all the simulations and data analysis, and wrote the original draft. YongQuan Wu designed the overall framework, proposed the key concepts, and revised the manuscript. JingLin You checked the design of all figures in the manuscript. All authors reviewed and approved the final version of the manuscript.

Competing interests

The authors declare no competing interests.

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